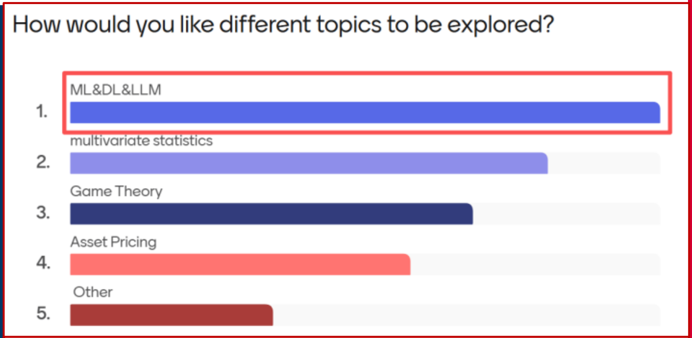
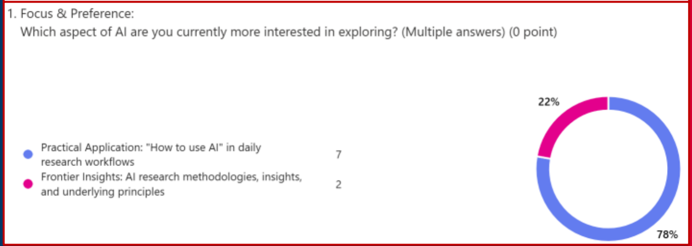
UEBS Statistics Study Group

UNIVERSITY OF EDINBURGH
Business School

AI / ML / DL in Business Research





Discussion-Based Workshop · Session 3

Moderator: Yifan Qi

Friday, 13 March 2026 | 17:00–18:00

University of Edinburgh Business School

WELCOME & CONTEXT

How We Got Here

5 min

1

23 Jan 2026

Kick-off Session

Decided to launch AI/ML/DL discussion-based workshop series as part of UEBS Stats Study Group

2

Session 1 & 2

Zhaoxi Zhang & Han Zhang's Talks

Shared their own research. Thank you to Zhaoxi Zhang & Han Zhang for contributing! 👏

- 13th Feb 2026 Zhaoxi Zhang: The generalized underlap coefficient with an application in clustering
- 27th Feb 2026 Han Zhang: The “sponsored content residue” in influencer marketing

3

Today

Session 3: Panorama

Structured overview — from AI concepts to research workflows to frontier paradigm shifts in business disciplines

Today's Goal: Build a shared mental model of AI in business research — concepts, tools, and frontier insights — then discuss where we go next as a group.

TODAY'S AGENDA

40-60 Minutes, Five Blocks

5 min

Welcome & Context

Kick-off recap, thank previous speakers

15-30 min

AI Panorama: Three Layers

Concepts → Workflow → Paradigm Shifts + guided discussion

10 min

Future Symposium Design

Video clips, website, resource sharing & legacy building

5-10 min

Upcoming Sessions & Sign-ups

27 Mar · 10 Apr · 24 Apr — topics, collaborations, mock presentations

5 min

Student Experience Grant

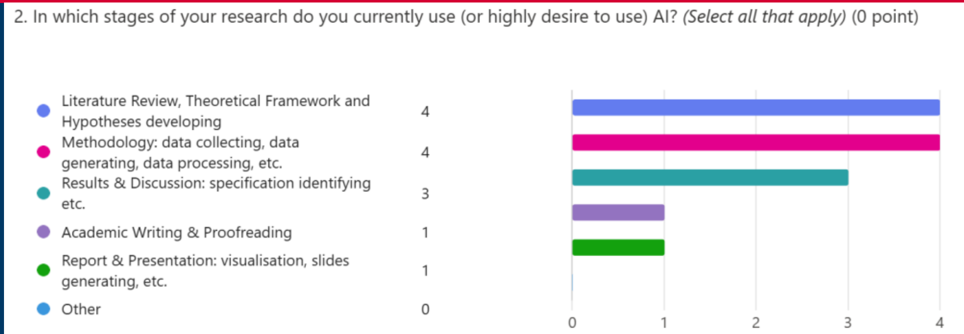
£900 fund — food, activities, community building

PART 1

The AI/ML/DL Panorama

Three layers: from foundational concepts to research workflows to discipline-specific paradigm shifts

10 min presentation + 10–15 min guided discussion



3. What specific AI-related knowledge, skill, or tool are you most eager to learn right now?

7 Responses

ID ↑	Name	Responses
1	anonymous	Python operation for data processing
2	anonymous	Codex, Claude Code and OpenClaw
3	anonymous	large language model and using it for data generation
4	anonymous	Agent
5	anonymous	Claude code
6	anonymous	none
7	anonymous	Skills

From AI to Agents: A Structured Map

Discussion: Which of these layers is most relevant to your current research? Where do you feel the biggest knowledge gap?

AI (Artificial Intelligence)

The umbrella — making machines exhibit intelligent behavior

→

Core Subfields

NLP · Computer Vision

Parallel fields solving language and image

Take away:
*NLP (1. **Data Stucturing**: 1) Classification, 2) topic modeling (identifying), 3) NER(Named Entity Recognition), 4) relation extraction; 2. **Quantification**(text to numbers and vectors) 3. **Semantic analysis**, 4) **Generation***

Data-Driven (ML)

Machine Learning → Deep Learning

From statistical pattern recognition to multi-layer neural networks

→

Deep Tech (DL+)

Transformers → LLMs → Agents

Self-attention architecture → GPT/Claude → autonomous tool-using

Take away:
LLM: ChatGPT, Claude, Gemini, Doubao, Deepseek, Qwen etc.

The Practical Spectrum: Prompting → RAG → Fine-tuning

Discussion: Have you tried RAG or fine-tuning? What stopped you — cost, complexity, or just not knowing where to start?

Prompting

80%

of use cases

Instructions in input.

No model changes.

Cost: near-zero.

Time: minutes.

Start here for all research tasks

RAG (Retrieval-Augmented Generation)

15%

of use cases

Retrieve your docs,

inject into context.

Cost: low–moderate.

Time: days.

When the model needs your specific papers

Fine-tuning

5%

of use cases

Further train on

task-specific data.

Cost: \$100–\$10K.

Time: days–weeks.

When consistency and domain expertise matter

Take away:
RAG is now getting embedded into the Agent:

```
├── Planning
├── Tool Use
│   ├── Search
│   ├── Code Execution
│   └── RAG Retrieval
└── Memory
```

AI Across Every Stage of Your PhD

Discussion: *Which workflow stage takes you the most time? What tools do you already use vs. want to learn? Would a hands-on "connect AI to Stata/R/Python" session be useful for a future workshop?*

Lit Review & Synthesis

ResearchRabbit, Elicit (99.4% acc), Scite.ai, Claude RAG

Data Collection & NLP

FinBERT (88% acc), LLM scrapers, SEC EDGAR parsers, S&P NLP

Econometrics & Stats Code

Stata-MCP, EconML, DoubleML, grf, GitHub Copilot

Experiment Design & Writing

SDV (synthetic data), Zotero+AI, Writefull, Overleaf sync

Take away:

Statistic coding and analysis: *Integrate AI with statistical software such as Stata and Python.*

Agent can deal with them automatically: *Openclaw , Gemini CLI, Claude Code*

Instruction of deploying Claude Code or Agent*:* *https://anthropic.skilljar.com/claude-code-in-action; https://anthropic.skilljar.com/introduction-to-agent-skills*

UOE ELM API (last year around $500): https://elm.edina.ac.uk/;

Other commercial API: Chatgpt, Claude, Gemini, Moonshot(Kimi); MiniMax; DeepSeek; Qwen; Zhipu; Doubao

How AI Reshapes Business Research

Accounting & Finance

- ML reshapes asset pricing (Gu et al. 2020)
- FinBERT > dictionaries for sentiment
- Satellite + DL for climate risk
- DML for causal policy effects
- LLMs as simulated traders

Mgmt Sci & Economics

- DML & Causal Forests become standard
- LLM agent-based economic simulations
- Text-as-data framework (Ludwig et al. 2025)
- Adaptive experimental design (bandits)
- Algorithmic decision-making impacts

Marketing

- CatBoost/XGBoost for consumer prediction
- Causal inference for personalization
- AI-generated experimental stimuli
- "AI-Authorship Effect" (Givi 2024)
- Synthetic respondents for piloting

Discussion: *Which paradigm shift is most exciting for your field? Is there a paper from the interactive map you'd like to explore together?*

UPCOMING SESSIONS

Three More Sessions — Your Turn to Shape Them

10 min

Fri 27 Mar
17:00–18:00
Runzhi Tian (Python) & Yifan Qi (Stata)

Fri 10 Apr
17:00–18:00

Fri 24 Apr
17:00–18:00

What Could You Present or Co-Create?

Hands-on Workshop

Teach the group **how to connect AI to Stata/R/Python**. Live coding, real datasets, practical tips.

Frontier Literature Insights

Share a paper from your reading that changed how you think about your research methodology.

Mock Presentations

Practice your Annual Review presentation, conference talk, or viva defense. Get constructive feedback.

Collaborative Knowledge Exchange Projects

Team up to build a shared resource — a code library, literature synthesis, or methodology guide.

Please sign up for topics you're interested in — we'll coordinate via group chat. Solo or collaborative sessions both welcome!

STUDENT EXPERIENCE GRANT

£900 Fund — How Should We Spend It?

5 min

£900

Student Experience Grant
from UEBS

Your Student Experience Grant Ap...

Project title: Business School Statistics Studying Group

Dear Yifan,

We have much pleasure in informing you that your project has been awarded a Student Experience Grant. The value of the grant to be awarded is £900 .

THE UNIVERSITY of EDINBURGH

Student Experience Grants

THE UNIVERS
ACCEPTANCE OF ST
This form must be returned within three

Grant ID:	GR005694
Name:	Yifan Qi
Grant awarded:	£900
Round awarded:	Autumn 202
College or department	College of Arts, Humanities and Social Sciences

Terms and conditions

By returning this form to the Grants team student-experience-grants@ed.ac.uk, I understand the following conditions of the grant and agree to fulfil them.

- I will NOT use the Student Experience Grant for any purpose other than that stated in my application or in the award conditions. Please note that the use of the grant for other purposes may result in repayment of the grant.
- I will acknowledge the support of alumni and friends of the University of Edinburgh where possible, including on equipment, websites, printed materials or event marketing. The Student Experience Grants logo and acknowledgement line are available here: https://student-experience-grants.ed.ac.uk/information-for-grant-recipients.
- I will return the grant report within one month of completion of my project to the Grants team.
- I will maintain a record of expenditure with receipts and submit them with the grant report.

Ideas on the Table

- Post-session dinners / coffee to build community and attract new members
- Invite external speakers (industry practitioners, other departments)
- Workshop materials — printed guides, cheat sheets, poster printing
- End-of-term symposium event with catering and networking

Budget item 1

Description of item 1

Light refreshments (biscuits, tea/coffee, soft drinks, small cakes) for seven biweekly study sessions

ts.ed.ac.uk/form/student-experience-grants-applic

Student Experience Grants Application | Student Experience Grants

(15–20 participants each). (£70 * 7 sessions & £90 * 1 Welcome Event ≈ £580 + £20 contingency for minor price variation)

Amount of item 1

£600

Do you wish to add a 2nd Budget Item?

Yes

Budget item 2

Description of item 2

End-of-semester summary session (pizza dinner) to celebrate completion, encourage reflection, and strengthen PGR community cohesion. (Approx. £20 × 15 participants)

Amount of item 2

£300

Discussion:
How should we split the budget across activities? Any creative ideas?

Take away:
Snacks: *Milk Tea, coke, Pizza, Chicken wings, refreshing veggies;*
Restaurant: *Wuzhen/Changan Chinese Restaurant; Buffet*

PART 2

Building a Symposium (will be discussed in next session)

Turning our workshop series into a lasting resource
for the UEBS PhD community

15 min group discussion

SYMPOSIUM DESIGN

From Workshop to Legacy

Video Documentation

Record talks, create short highlight reels for the UEBS community. Future PhD students can learn from our discussions.

Website / Resource Hub

A shared page hosting slides, code snippets, the interactive map, and curated reading lists organized by discipline.

Knowledge Transfer

Share existing resources — datasets, Stata/Python code, literature maps — so knowledge compounds instead of disappearing.

Cross-Cohort Continuity

Design the symposium so new PhD/PGR members can pick it up. Templates, documented workflows, and a shared drive.

Discussion: *Which format would be most valuable? Would you contribute a short video or write-up?*
What resources do you wish existed when you started your PhD?

Thank You!

Next Steps

Explore the interactive panorama — click nodes, read the Core Insights

Sign up for a topic slot (10 Apr · 24 Apr)

Share your ideas for the symposium format via group chat

Vote on how to spend the £900 Student Experience Grant
What snacks&beverage(drinks) and restaurant you like?

Interactive Panorama: HTML file shared via group chat | GitHub link forthcoming
All slides and the detailed Markdown framework are available for download